

A Review On Feature Selection For High Dimensional Data

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Abstract—Feature selection is very important as data is created constantly and at an ever increasing rate, it helps to reduce the high dimensionality of some problems. Feature selection as a preprocessing step to machine learning, is effective in reducing redundancy, removing irrelevant data, increasing learning accuracy, and improving result comprehensibility. This work offers comprehensive approach to feature selection in the scope of classification problems, explaining the foundations, real application problems etc in the context of high dimensional data. First, we focus on the basis of feature selection provides an analysis on history and basic concepts. The different types of feature selection methods are discussed and finally analyze the findings.

I. INTRODUCTION

We are in a data-driven era, in which the size of digital data is continuously growing since data are acquired for many purposes. Hence, machine learning algorithms have to cope with data sets whose volume and complexity are also increasing. Feature selection is the process of electing the attributes that allow to clearly define a problem, apart from those that are irrelevant or redundant. Feature selection methods that, traditionally, can be categorized into three groups filters, wrappers and embedded. In addition, many new methods combining existing methods or other machine learning techniques are continuously appearing to address the challenges of today's datasets.

A. Need for data selection

As Internet has been growing at unprecedented rates, lots of data come in different formats text, multimedia, etc from different sources systems, sensors, mobile devices, etc. They require new tools and technology in order to be analyzed and processed so as to enable the extraction of useful information. The ultra high dimensionality leads to unbearable memory requirements and high computational cost in training. Any dataset can be represented by a matrix where the rows are the samples or instances that have been recorded and the columns are the attributes or features that are needed to represent the problem. Therefore need to find the narrower matrices to tackle the curse of dimensionality problem. These narrow matrices have only a small number of samples and a small number of attributes, and therefore can be used much more efficiently. The process of finding these narrow matrices is called dimensionality reduction.

Feature extraction is one of the techniques used for dimensionality reduction, which finds the most compact and informative set of features of a given problem, to improve the efficiency or data storage or processing.

B. Problems that feature selection has to deal with

There are numerous problems that can benefit from the application of feature selection techniques.

- Small Sample Size

High dimension, low sample size (HDLSS) data are becoming increasingly common in various fields. Many of the features of these problems do not facilitate an adequate classification. To avoid this problem feature selection have been proposed.

- Class imbalance problem

Imbalance problem occur when one of the two classes having more samples than other classes. The most of algorithm ignore the minority sample while focus on classification of major sample. The minority samples are very important however they are rarely occur but very important. Among different solutions proposed for this problem, few are data sampling, algorithmic modification, cost sensitive learning and feature selection.

- Dataset Shift

Dataset shift is a common problem in machine learning that occurs when the joint distribution of inputs and outputs differs between training and test stages. Covariate shift, a particular case of dataset shift, occurs when only the input distribution changes. Feature selection helps to minimizes this problem.

- Noisy Data

Dimensionality reduction and consequently feature selection is one of the most popular techniques to remove noisy data[1].

- Feature Cost

Test costs and misclassification costs are the two most important types of cost in cost-sensitive learning.

Feature selection works have mainly considered misclassification costs[2].

II. FEATURE SELECTION METHODS

The feature selection methods are typically presented in three classes based on how they combine the selection algorithm and the model building.

- Filters, which rely on the general attributes of training data and carry out the feature selection process as a preprocessing step with independence of the induction algorithm. Advantageous of this model are its low computational cost and good generalization ability.

Set of features → Selecting best subset → Learning algorithm → Performance

- Wrappers, assess the relative usefulness of subsets of variables which allows detecting interaction between variables. Steps involves in wrapper method is shown in Fig1. Disadvantages of this method are over fitting risk when observations are insufficient and significant computation time when the number of variables is large.

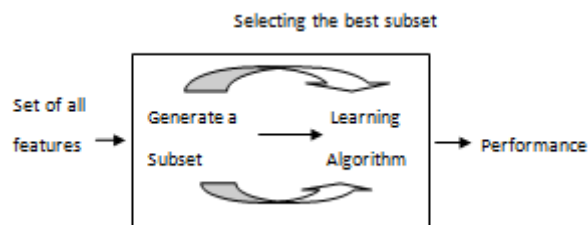


Fig1: Wrapper method for feature selection

- Embedded methods, which combines the advantages of filters and wrappers. This approach takes advantage of its own variable selection process and performs feature selection and classification simultaneously. Steps involves in the embedded method is shown in Fig2.

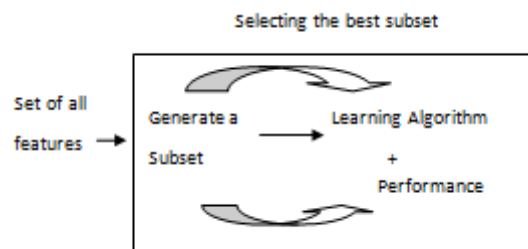


Fig2: Embedded method for feature selection

A. Filter methods

Filter methods are based on performance evaluation metrics calculated directly from the data, without direct feedback from predictors that will finally be used on data with a reduced number of features [3]. These algorithms are normally computationally less expensive than wrappers or embedded methods.

- Chi-Squared

This is an univariate filter based on the χ^2 statistic [5]. It evaluates each feature independently with respect to the classes. The higher the value of chi-squared, the more relevant is the feature with respect to the class.

- Information Gain

The Information Gain filter [6] is one of the most common univariate methods of evaluation attributes. In this features are evaluated according to their information gain and consider a single feature at a time. The entropy is considered as a measure to rank variables. The entropy of a class feature Y is

$$H(Y) = -\sum p(y) \log_2(p(y)),$$

where $p(y)$ is the marginal probability density function for the random variable Y. Then the entropy of Y after observing X is

$H(Y|X) = -\sum p(x)p(y|x) \log_2(p(y|x))$, where $p(y|x)$ is the conditional probability of y given x. Given the entropy as a criterion of impurity in a training set S, we can define a measure reflecting additional information about Y provided by X that represents the amount by which the entropy of Y decreases. This measure, an indicator of the dependency between X and Y, is known as IG:

$$IG = H(Y) - H(Y|X) = H(X) - H(X|Y)$$

IG is a symmetrical measure. A weakness of the IG criterion is that it is biased towards the features with more values even when they are not more informative.

- Fast Correlation-Based Filter, FCBF

The fast correlated-based filter method [11] is a multivariate algorithm that measures feature class and feature-feature correlation. FCBF selects a set of features that is highly correlated with the class based on symmetrical uncertainty (SU). SU is defined as the ratio between the information gain and the entropy of two features. Then, it applies three heuristics that remove the redundant features and keep those features that are more relevant to the class. FCBF was designed mainly for high-dimensionality data and is effective in removing both irrelevant and redundant features.

- INTERACT

The INTERACT algorithm [7] uses the same goodness measure as the FCBF filter, i.e., SU, but it also includes the consistency contribution, which is an indicator of how significantly the elimination of a feature will affect consistency. The algorithm consists of two major parts. In the first part, the features are ranked in descending order based on their SU values. In the second part, features are evaluated one by one starting from the end of the ranked list of feature. If the consistency contribution of a feature is less than an established threshold, the feature is removed; otherwise it is selected.

- A Fast Clustering-Based Feature Subset Selection

The FAST[8] algorithm works in two steps. In the first step, features are divided into clusters by using graph-theoretic clustering methods. In the second step, the most representative feature that is strongly related to target classes is selected from each cluster to form a subset of features. Features in different clusters are relatively independent, the clustering-based strategy of FAST has a high probability of producing a subset of useful and independent features. To ensure the efficiency of FAST, uses the efficient minimum-spanning tree (MST) clustering method.

B. Wrapper method

The wrapper approach[4] is to select a feature subset using a learning algorithm based on evaluation function. Instead of using subset sufficiency, entropy or another explicitly defined evaluation function can be used to guide the search. The evaluation function for each candidate feature subset returns an estimate of the quality of the model that is induced by the learning algorithm. As each candidate feature subset evaluated during the search, the target learning algorithm need to be applied several times. Wrapper methods are classified into Sequential Selection Algorithms and Heuristic Search Algorithms.

- Sequential selection algorithms

These algorithms are called sequential because it is iterative in nature. The Sequential Feature Selection (SFS) algorithm starts with an empty set and adds the feature that has highest objective function. Iteratively remaining features are added to current subset and new subset is evaluated. The feature is permanently included in the subset if it gives the maximum classification accuracy. This is a naive SFS algorithm as it is not considered the relation between the features. A Sequential Backward Selection (SBS) algorithm can also be constructed; this algorithm starts from the complete set of variables and removes one feature at a time. Remove those features whose removal gives the lowest decrease in predictor performance.

The Sequential Floating Forward Selection[12] (SFFS) algorithm is more flexible than the naive SFS. The first step of the algorithm is the same as the SFS algorithm which adds one feature at a time based on the objective function. In SFFS algorithm additional step is excluding one feature at a time from the subset obtained in the first step and evaluates the new subsets. If this process increases the value of the objective function then that feature is removed and goes back to the first step with the new reduced subset or else the algorithm is repeated from the top. Repeat above steps until the required number of features is added or required performance is reached.

- Heuristic search algorithms

Genetic Algorithm (GA) can be used to find the subset of features. The chromosome bits represent if the feature is included or not. The global maximum for the

objective function can be found which gives the best suboptimal subset. The GA parameters and operators can be modified within the general idea of an evolutionary algorithm to suit the data or the application to obtain the best performance or the best search result.

C. Embedded methods

Embedded methods do not separate the learning from the feature selection part. This method include algorithms, which optimize a regularized risk function with respect to two sets of parameters: the parameters of the learning machine and the parameters indicating which features are selected. The embedded methods used are subsequently described.

- Recursive Feature Elimination for Support Vector Machines SVM-RFE

This embedded method was introduced by Guyon in [9]. It performs feature selection by iteratively training an SVM classifier with the current set of features and removing the least important feature as indicated by the weights in the SVM solution.

- Feature Selection-Perceptron, FS-P

FS-P [10] is an embedded method based on a perceptron. A perceptron is a type of artificial neural network that can be seen as the simplest kind of feed forward neural network: a linear classifier. The basic idea of this method consists in training a perceptron in the context of supervised learning. The interconnection weights are used as indicators of which features could be the most relevant and provide a ranking.

III. RESULTS

Feature selection has been an active and fruitful field of research in machine learning. Its importance is indisputable and it has proven effective in increasing predictive accuracy and helps to reduce complexity of machine learning models. For this reason, new feature selection methods have been emerging during the past few decades. The feature selection algorithms, allows intelligent selection from existing algorithms. There are few general rules of thumb, such as choosing filters when dealing with extremely large data sets and using embedded and wrappers when the computational burden is not an issue, but some level of expertise is necessary to choose a specific method. In the machine learning field, common factors that might affect the performance of a feature selection method, are proportion of irrelevant features present in the data or the interaction between attributes. Another important factor to take into account is the noise in the data.

In the current scenario, huge amounts of data are continuously generated, so it becomes more and more difficult to find different features. A common problem, while testing the effectiveness of feature selection methods is that it is not always possible to know the relevant features a priori, so their performance depends

on performance of the learning method used afterwards and it can vary notably from one method to another.

Table 1 shows comparison of few feature selection methods.

Method	Uni/multi variate	Fun ctional view	Structural view	Complexity
Chi-squared	Univariate	Ranker	Filter	nm
Information gain	Univariate	Ranker	Filter	nm
FCBF	Multivariate	Subset	Filter	nmlogn
FAST	Multivariate	Subset	Filter	n^2
INTERACT	Multivariate	Subset	Filter	nm^2

Average of success for every feature selection methods are

- When dealing with high nonlinearity in data, SVM-RFE with a Gaussian kernel is an excellent choice, since it is able to solve these complex problems, however, at the expense of being computationally more expensive than the remaining approaches.
- In the presence of altered inputs, the best option is to use the embedded method FS-P, since it has proved to be very robust to noise.
- In general, the use of filters, since they carry out the feature selection process independently of the induction algorithm and are faster than embedded and wrapper methods. However, in case of using another approach, suggest using the embedded method FS-P.

IV. CONCLUSION

However feature selection has been applied for decades, its importance is increasing not only because of huge volume of data getting generated created but also with other aspects such as such as class imbalance problem, data complexity, dataset shift, non-stationary distributions, etc which open new lines of research in which the use of feature selection is, perhaps, more necessary than ever. This new scenario offers both challenges and opportunities to machine learning researchers, who should embrace this opportunity to launch new lines of research. From time complexity and performance analysis it has been seen that FAST and FCBF gives better efficiency in feature selection.

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